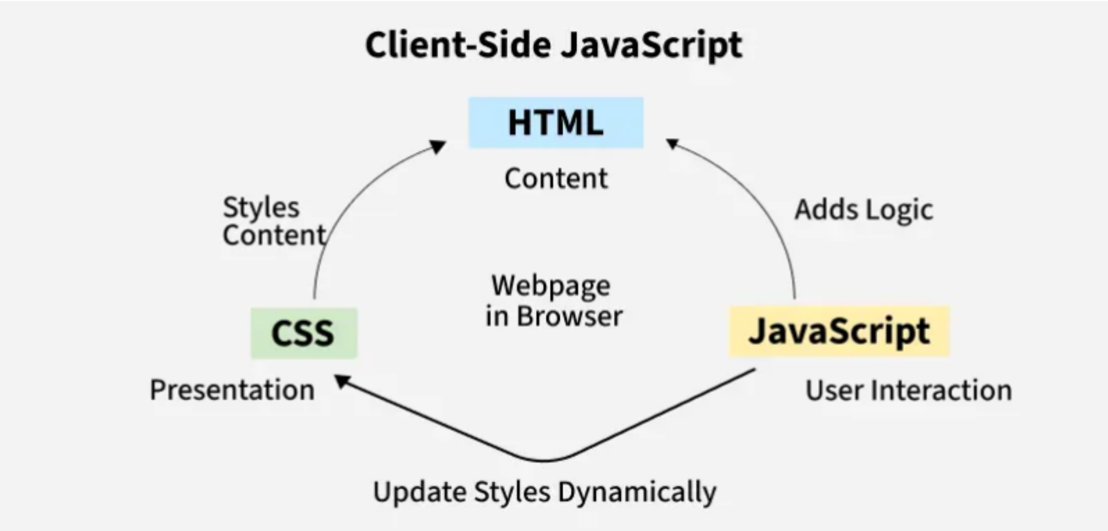
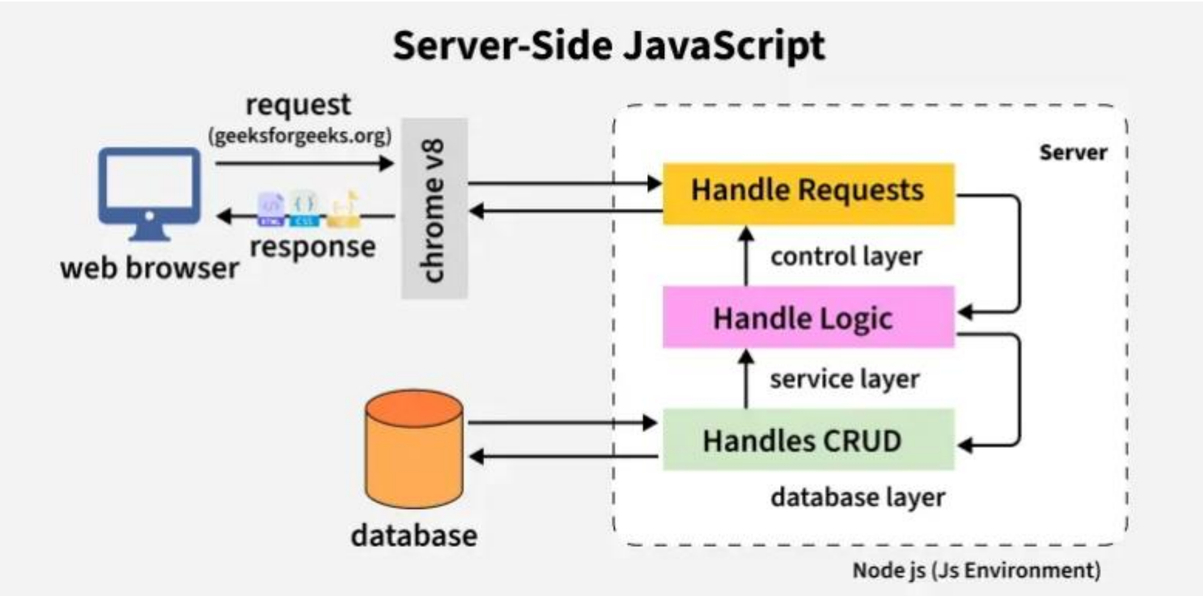
JavaScript

JavaScript is a programming language used to create dynamic content for websites. It is a lightweight, cross-platform, and single-threaded programming language. It's an interpreted language that executes code line by line, providing more flexibility.

JavaScript on Client Side: On the client side, JavaScript works along with HTML and CSS. HTML adds structure to a web page, CSS styles it, and JavaScript brings it to life by allowing users to interact with elements on the page, such as actions on clicking buttons, filling out forms, and showing animations. JavaScript on the client side is directly executed in the user's browser.



JavaScript on Server Side: On the Server side (on Web Servers), JavaScript is used to access databases, file handling, and security features to send responses, to browsers.



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Why JavaScript?

- JavaScript is core language for web development, enabling dynamic and interactive features in websites with fewer lines of code.
- It is highly in demand, offering many job opportunities in Frontend, Backend (Node.js), and Full Stack Development.
- JavaScript supports powerful frameworks and libraries like React, Angular, Vue.js, Node.js, and Express.js, widely used in modern web applications.
- JavaScript is an object-oriented and event-driven language, ideal for building scalable and responsive applications.
- It is cross-platform and runs directly in all modern web browsers without the need for installation.
- Major companies like Google, Facebook, and Amazon use JavaScript in their tech stacks.

Basic JavaScript Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Run JS</title>
</head>
<body>

<script>
  let a = 5;
  let b = 6;
  let c = a + b;
  document.writeln(c);
</script>

</body>
</html>
```

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"JavaScript can update and change both HTML and CSS, it also can calculate, manipulate and validate data" – Justify the statement with coding.

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Javascript Understanding..</title>
    <link rel="stylesheet" href="style.css">
    <script src="script.js"></script>
```

```html
</head>
<body>
    <input type="text" id = "num1" placeholder="Enter 1st Number">
    <input type="text" id = "num2" placeholder="Enter 2nd Number">
    <button onclick="calculate()">ADD</button>

    <p id="message">Result will Show here..</p>
</body>
</html>
```

Style.css

```css
#message{
    color: green;
}

.error{
    color: red !important;
}
```

Script.js

```javascript
function calculate(){
    var a = document.getElementById("num1").value;
    var b = document.getElementById("num2").value;

    if(isNaN(a)||isNaN(b)||a===""||b==="")
    {
        document.getElementById("message").innerHTML="Invalid number";
        document.getElementById("message").className="error";
    }
    else{
        var sum = Number(a)+Number(b);
        document.getElementById("message").innerHTML="The Sum is = " + sum;
        document.getElementById("message").className = "";
    }
}
```

Update & Change HTML: JavaScript uses innerHTML to dynamically show messages and results inside the <p id="message"> element.

Update & Change CSS: JavaScript changes the text color by adding the .error class using className, which updates the CSS style.

Calculate & Manipulate Data: It takes numeric input, converts strings to numbers with Number(), and adds them using +.

Validate Data: It checks if the input is a number or empty using isNaN() and a=== "", b=== "", ensuring valid data before calculation.